INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

THRUSTER, CARTRIDGE ACTUATED, M1A2 METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M1A2 Cartridge Actuated Thruster Metal Parts Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-2550	Ammunition, General Specification For

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-413	Visual Inspection Guide for Elastomeric O-rings

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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(Unless otherwise indicated, copies of federal and military specifications and standards are available from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.1.2 Other Government documents, drawings, and publications. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

Naval Surface Warfare Center, Indian Head (CAGE Codes 19200 and 19203)

8593855	Thruster, Cartridge Actuated, M1A2 Metal Parts Assembly
8797631	Box, Fiberboard (Inner), Packing, Ammunition, For Thruster, Cartridge Actuated, M1A2
8797632	Box, Packing, Ammunition, For Thruster, Cartridge Actuated, M1A2
9221261	Box, Fiberboard (Outer), Alternative, Packing Ammunition, for Thruster, Cartridge Actuated, M1A2

U.S. Air Force

TPO 00-891-5488	Thruster, Cartridge Actuated, M1A2

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Data Control Branch (Code 8410), Indian Head, MD 20640-5035.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to the first article inspection (see 4.3 and 6.3).

3.2 Materials. All materials used in the manufacture of the thrusters shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2.1 Raw material. The contractor shall be responsible for selecting from the classification list contained in the raw material specification the surface and physical condition of the raw material intended for use in manufacture.

3.3 Hardness and other physical properties. Hardness and other physical properties required by the drawing shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

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3.4 Hydrostatic pressure. The body and head shall withstand a hydrostatic pressure of 7,500 pounds per square inch (psi) for 15 seconds, minimum, without leaking, deformation, or mechanical failure (see 6.4.1).

3.5 Shear pin strength. The shear pin, after application of protective finish, shall shear when a load of 48 ± 4 pounds is applied at the steady rate of 0.125 inch per minute.

3.6 Metal parts assembly. Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing. All threaded joints shall be assembled handtight.

3.7 Protective finish. Protective finish shall be in accordance with the applicable specifications and shall apply to all surfaces of the component regardless of contour configuration.

3.8 Residual magnetism. The metal parts assembly shall not deflect the indicator of the compass more than five degrees in either direction using procedures in 4.6.5.

3.9 Workmanship. Thrusters shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.9.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.9.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.9.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.3 Inspections. Inspections shall be in accordance with MIL-A-2550.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: room ambient (68 to 77•F)
b. Altitude: sea level
c. Humidity: 50 percent relative, maximum
d. Atmospheric pressure: 14.7 psi

4.4 First article inspection. At the beginning of regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of five metal parts assemblies, nine sets of unassembled parts, one set of unassembled parts without plating or final protective finish, and fifty shear pins from the same lot of shear pins tested by the contractor. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, must be the same as will be used for regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the first article sample shall be inspected for all requirements of the drawings and specifications. The Government reserves the right to require new first article samples until an acceptable sample is submitted (see 6.3).

4.5 Quality conformance inspection.

4.5.1 Lot.

4.5.1.1 Submission of product. Unless otherwise specified by the contracting activity, formation and identification of lots shall be in accordance with MIL-STD-105.

4.5.1.2 Size of lot. The production lot size shall be as specified in the contract.

4.5.2 Examination. One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects. Four power (4x) magnification may be used in performing visual examination for such defects as burr and foreign matter. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

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4.5.2.1 Key, Drawing 8593857, covering a detail of Drawing 8593855.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Radial profile of length, inside and outside	Gage
102	Radial profile of width	Gage
103	Surface finish incorrect	Visual
104	Protective finish inadequate or defective	Visual
105	Metal defective	Visual
106	Burr or sharp corner	Visual
107	Foreign matter	Visual
Minor		
201	Total length	Gage
202	Radius or chamfer incorrect	Gage

4.5.2.2 Pin, Firing, Drawing 8593864, covering a detail of Drawing 8593855.

Critical		
None defined		
Major		
101	Concentricity of groove with body	Gage
102	Concentricity of striker with body	Gage
103	Concentricity of striker shoulder with body	Gage
104	True position of hole with body	Gage
105	Diameter of groove	Gage
106	Diameter of body, min	Gage
107	Diameter of striker, max	Gage
108	Diameter of pin hole, max	Gage
109	Depth of pin hole	Gage
110	Profile of striker	Gage
111	Profile of groove	Gage
112	Metal defective	Visual
113	Protective finish inadequate or defective	Visual
114	Surface finish of body and striker tip incorrect	Visual
115	Burr or sharp corner	Visual
116	Foreign matter	Visual
Minor		
201	Diameter of body, max	Gage
202	Diameter of striker, min	Gage
203	Diameter of pin hole, min	Gage
204	Diameter of striker shoulder	Gage
205	Depth of pin hole	Gage

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206	Length from body to tip of striker	Gage
207	Length of striker	Gage
208	Total length	Gage
209	Location of pin hole	Gage
210	Location of groove	Gage
211	Depth of slot in base	Gage
212	Length of body diameter forward of groove, min	Gage
213	Width of groove	Gage
214	Width of slot in base	Gage
215	Location of slot in base	Gage
216	Radius incorrect	Gage
217	Radius missing	Visual
218	Surface finish improper, except as otherwise classified	Visual
219	Marking missing, incorrect, or illegible	Visual

4.5.2.3 Body Assembly, Drawing 8593859, covering a detail of Drawing 8593855.

Critical
None defined

Major

101	Pitch diameter of thread, min	Gage
102	Major diameter of thread, min	Gage
103	Concentricity of pitch diameter of thread with large bore	Gage
104	Concentricity of small bore with large bore	Gage
105	Concentricity of groove with small bore	Gage
106	Concentricity of body, thread end, with pitch diameter of thread	Gage
107	Concentricity of counterbore with small bore	Gage
108	Perpendicularity of face at threaded end with pitch diameter of thread	Gage
109	Perpendicularity of face of flange with pitch diameter of thread	Gage
110	Diameter of large bore, max	Gage
111	Diameter of counterbore	Gage
112	Diameter of key groove	Gage
113	Diameter of small bore	Gage
114	Diameter of body, thread end	Gage
115	Diameter of internal undercut, max, if applicable	Gage
116	Location of key groove	Gage
117	Straightness of bore	Gage
118	Diameter of thread undercut, min	Gage
119	Profile of key groove	Gage
120	Thread damaged or not full	Visual
121	Metal defective	Visual
122	Protective finish inadequate or defective	Visual
123	Surface finish of large bore improper	Visual
124	Burr or sharp corners	Visual
125	Foreign matter	Visual

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Minor

201	Pitch diameter of thread, max	Gage
202	Diameter of large bore, min	Gage
203	Diameter of body, max	Gage
204	Depth of counterbore	Gage
205	Total length	Gage
206	Length of flange	Gage
207	Thickness of wall	Gage
208	Diameter of flange	Gage
209	Length to start of angle at flange, min	Gage
210	Length to end of angle at flange, max	Gage
211	Length to large bore	Gage
212	Length to body, thread end	Gage
213	Width of thread undercut	Gage
214	Width of internal undercut, max, if applicable	Gage
215	Width of flats	Gage
216	Width across flats	Gage
217	Location of flats	Visual
218	Marking missing, incorrect, or illegible	Visual
219	Radius or chamfer incorrect	Gage
220	Surface finish improper, except as otherwise classified	Visual

4.5.2.4 Body Assembly, Drawing 8593859, covering a detail of Drawing 8593855.

Critical
None defined

Major

101	Diameter across top of pellets, max	Gage
102	Protrusion of pellet above major diameter of thread, min	Gage
103	Pellet missing, loose, or improperly inserted	Manual/Visual
104	Thread damaged or not full	Visual
105	Metal defective	Visual
106	Burr	Visual
107	Foreign matter	Visual

Minor

201	Location of pellet	Gage
202	Pellet radius incorrect	Visual

4.5.2.5 Trunnion, Drawing 8593860, covering a detail of Drawing 8593855.

Critical
None defined

Major

101	Concentricity of lugs	Gage
102	Symmetry of lugs with inside diameter	Gage

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103	Symmetry of sides with inside diameter	Gage
104	Symmetry of flats at root of lugs with inside diameter	Gage
105	Diameter of lugs, min	Gage
106	Width	Gage
107	Width across flats, min	Gage
108	Thickness of wall, min	Gage
109	Radius at base of lugs incorrect	Gage
110	Metal defective	Visual
111	Protective finish inadequate or defective	Visual

Minor

201	Diameter of lugs, max	Gage
202	Diameter, inside	Gage
203	Length of lugs	Gage
204	Width across flats, max	Gage
205	Thickness of wall, max	Gage
206	Location of lugs	Gage
207	Marking missing, incorrect, or illegible	Visual
208	Radius or chamfer missing or incorrect, except as otherwise classified	Visual
209	Surface finish improper	Visual
210	Burr or sharp corner	Visual
211	Foreign matter	Visual

4.5.2.6 Retainer, Cartridge, Assembly, Drawing 8593862, covering a detail of Drawing 8593855 (before inserting pellet).

Critical
None defined

Major

101	Pitch diameter of thread, min	Gage
102	Major diameter of thread, min	Gage
103	Concentricity of groove with flange	Gage
104	Concentricity of small cavity with large cavity	Gage
105	Concentricity of flange with pitch diameter of threads	Gage
106	Concentricity of large cavity with pitch diameter of threads	Gage
107	Diameter of flange, min	Gage
108	Diameter of groove	Gage
109	Diameter of large cavity, max	Gage
110	Diameter of small cavity, max	Gage
111	Internal radius	Gage
112	Length of effective thread, min	Gage
113	Profile of groove	Gage
114	Number of elongated slots incorrect	Visual
115	Escapement hole at rear of flange missing	Visual
116	Thread damaged or not full	Visual
117	Metal defective	Visual
118	Protective finish inadequate or defective	Visual

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119	Burr or sharp corner	Visual
120	Foreign matter	Visual
Minor		
201	Pitch diameter of thread, max	Gage
202	Diameter of flange, max	Gage
203	Diameter of large cavity, min	Gage
204	Diameter of small cavity, min	Gage
205	Diameter of rear of thread	Gage
206	Diameter of small end	Gage
207	Depth of large cavity	Gage
208	Depth to bottom of small cavity	Gage
209	Width of elongated slot	Gage
210	Diameter of escapement hole	Gage
211	Depth of base slot	Gage
212	Total length	Gage
213	Length to flange	Gage
214	Length of elongated slot	Gage
215	Length to end of flange	Gage
216	Width of groove	Gage
217	Width of base slot	Gage
218	Location of groove	Gage
219	Location of elongated slot	Gage
220	Location of escapement hole	Gage
221	Location of base slot	Gage
222	Marking missing, incorrect, or illegible	Visual
223	Radius or chamfer incorrect, except as otherwise classified	Gage
224	Surface finish improper	Visual

4.5.2.7 Retainer, Cartridge, Assembly, Drawing 8593862, covering a detail of Drawing 8593855 (after inserting pellet).

Critical
None defined

Major		
101	Diameter across top of pellets, max	Gage
102	Protrusion of pellet above major diameter of thread, min	Gage
103	Pellet missing, loose, or improperly inserted	Manual/Visual
104	Thread damaged or not full	Visual
105	Metal defective	Visual
106	Burr	Visual
107	Foreign matter	Visual
Minor		
201	Location of pellet	Gage
202	Pellet radius incorrect	Visual

4.5.2.8 Guide, Firing Pin, Drawing 8593863, covering a detail of Drawing 8593855.

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Critical
None defined

Major

101	Concentricity of body with cavity	Gage
102	Concentricity of striker hole with cavity	Gage
103	Concentricity of boss cavity with cavity	Gage
104	Symmetry of pin hole in body with cavity	Gage
105	Diameter of cavity, max	Gage
106	Diameter of body, min	Gage
107	Diameter of datum location	Gage
108	Diameter of pin hole, max	Gage
109	Height of boss	Gage
110	Angle at datum diameter	Gage
111	Radius at root of boss	Gage
112	Escapement hole missing	Visual
113	Metal defective	Visual
114	Protective finish inadequate or defective	Visual
115	Surface finish of cavity improper	Visual
116	Burr or sharp corner	Visual
117	Foreign matter	Visual

Minor

201	Diameter of cavity, min	Gage
202	Diameter of body, max	Gage
203	Diameter of pin hole, min	Gage
204	Diameter of striker hole	Gage
205	Diameter of boss cavity	Gage
206	Depth of striker hole	Gage
207	Length from bottom of cavity to face of boss	Gage
208	Total length	Gage
209	Location of pin hole	Gage
210	Diameter of escapement hole	Gage
211	Location of escapement hole	Gage
212	Marking missing, incorrect, or illegible	Visual
213	Radius or chamfer missing, except as otherwise classified	Gage
214	Surface finish improper, except as otherwise classified	Visual

4.5.2.9 Piston and Plug Assembly, Drawing 8593866, covering a detail of Drawing 8593855.

Critical
None defined

Major

101	Diameter of body, large end	Gage
102	Metal defective	Visual
103	Protective finish inadequate or defective	Visual
104	Burr	Visual
105	Foreign matter	Visual

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Minor
None defined

4.5.2.10 Piston, Drawing 8593867, covering a detail of Drawing 8593855.

Critical
None defined

Major

101	Concentricity of ring groove with body, large end	Gage
102	Concentricity of body, small end with body, large end	Gage
103	Concentricity of counterbore with body, large end	Gage
104	Concentricity of midsection with body, large end	Gage
105	Concentricity of locking groove with body, small end	Gage
106	Concentricity of small cavity with body, small end	Gage
107	Concentricity of key escape groove with body, small end	Gage
108	Diameter of body, large end, min	Gage
109	Diameter of ring groove	Gage
110	Diameter of shoulder, large end, min	Gage
111	Diameter of body, midsection	Gage
112	Diameter of body, small end	Gage
113	Diameter of key escape groove, max	Gage
114	Diameter of key locking groove, min	Gage
115	Diameter of small cavity	Gage
116	Diameter of undercut, min, if applicable	Gage
117	Diameter of counterbore	Gage
118	Length to body, large end, min	Gage
119	Length to key locking groove, min	Gage
120	Length of shoulder, large end	Gage
121	Location of angle in key groove	Gage
122	Depth of large cavity	Gage
123	Length to end of key escape groove	Gage
124	Length of body, small end, min	Gage
125	Profile of key escape and locking groove	Gage
126	Profile of ring groove	Gage
127	Metal defective	Visual
128	Protective finish inadequate or defective	Visual
129	Surface finish on body, large end and ring groove, improper	Visual
130	Burr or sharp corner	Visual
131	Foreign matter	Visual

Minor

201	Diameter of body, large end, max	Gage
202	Diameter of shoulder, large end, max	Gage
203	Diameter of key escape groove, min	Gage
204	Diameter of key locking groove, max	Gage
205	Length to body, large end, max	Gage
206	Length to key locking groove, max	Gage
207	Diameter of large cavity	Gage

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208	Depth of counterbore	Gage
209	Depth of small cavity	Gage
210	Length to body, midsection, max	Gage
211	Total length	Gage
212	Width of ring groove	Gage
213	Width of undercut, max, if applicable	Gage
214	Location of ring groove	Gage
215	Thickness between cavities	Gage
216	Radius or chamfer incorrect	Gage
217	Surface finish improper, except as otherwise classified	Visual

4.5.2.11 Plug, Piston, Drawing 8593868, covering a detail of Drawing 8593855.

Critical
None defined

Major

101	Diameter	Gage
102	Metal defective	Visual

Minor

201	Thickness	Gage
202	Chamfer incorrect	Gage
203	Protective finish inadequate or defective	Visual
204	Surface finish improper	Visual
205	Burr or sharp corners	Visual
206	Foreign matter	Visual

4.5.2.12 Sleeve, End, Drawing 8593869, covering a detail of Drawing 8593855.

Critical
None defined

Major

101	Pitch diameter of external threads, min	Gage
102	Major diameter of external threads, min	Gage
103	Concentricity of body with cavity	Gage
104	Concentricity of shoulder with body	Gage
105	Symmetry of elongated slot with body	Gage
106	Diameter of body, min	Gage
107	Diameter of cavity, max	Gage
108	Diameter of shoulder, min	Gage
109	Width of slot	Gage
110	Location of slot	Gage
111	Radius of threaded end	Gage
112	Threads damaged or not full	Visual
113	Metal defective	Visual
114	Protective finish inadequate or defective	Visual
115	Burr or sharp corner	Visual

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116	Foreign matter	Visual
Minor		
201	Pitch diameter of external thread, max	Gage
202	Pitch diameter of internal thread	Gage
203	Minor diameter of internal thread, max	Gage
204	Diameter of body, max	Gage
205	Diameter of cavity, min	Gage
206	Diameter of shoulder, max	Gage
207	Diameter of flange	Gage
208	Diameter of thread undercut, min	Gage
209	Length of slot	Gage
210	Variation in alignment, flange face with face of cavity seat	Gage
211	Depth of effective internal threads, min	Gage
212	Depth of small threaded hole, max	Gage
213	Total length	Gage
214	Width across flats	Gage
215	Width of shoulder	Gage
216	Width of thread undercut	Gage
217	Thickness of flange	Gage
218	Thickness of flange at flats	Gage
219	Location of flange	Gage
220	Location of flats	Gage
221	Marking missing, incorrect, or illegible	Visual
222	Radius or chamfer incorrect, except as otherwise classified	Gage
223	Surface finish improper	Visual

4.5.2.13 Head, Drawing 8593870, covering a detail of Drawing 8593855.

Critical
None defined

Major		
101	Pitch diameter of small thread, max	Gage
102	Minor diameter of small thread, max	Gage
103	Pitch diameter of midsection thread, max	Gage
104	Minor diameter of midsection thread, max	Gage
105	Pitch diameter of large thread, max	Gage
106	Minor diameter of large thread, max	Gage
107	Concentricity of pitch diameter of midsection thread at specified length with pitch diameter of large thread and base of large cavity	Gage
108	Perpendicularity of base of large cavity with pitch diameter of large thread	Gage
109	Perpendicularity of base of midsection cavity with pitch diameter of large thread	Gage
110	Depth of counterbore, small end	Gage
111	Depth of cavity, small end	Gage
112	Length of effective thread, small end, min	Gage

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113	Length of thread, small end, max	Gage
114	Diameter of undercut, large thread, max	Gage
115	Diameter of undercut, midsection thread, max	Gage
116	Diameter of chamfer at small end	Gage
117	Width of undercut, large thread	Gage
118	Width of undercut, midsection thread	Gage
119	Angle of chamfer at small end	Gage
120	Thread damaged or not full	Visual
121	Metal defective	Visual
122	Protective finish inadequate or defective	Visual
123	Surface finish of base of midsection cavity, small thread face and chamfer of angle improper	Visual
124	Burr	Visual
125	Foreign matter	Visual

Minor

201	Pitch diameter of small thread, min	Gage
202	Pitch diameter of midsection thread, min	Gage
203	Pitch diameter of large thread, min	Gage
204	Diameter of counterbore, small end	Gage
205	Diameter of hole between midsection and small end	Gage
206	Depth of large cavity	Gage
207	Depth to bottom of midsection	Gage
208	Diameter of body midsection	Gage
209	Diameter of body, large end	Gage
210	Diameter of body, small end	Gage
211	Length of large body diameter	Gage
212	Length of small body diameter	Gage
213	Length of flat	Gage
214	Total length	Gage
215	Width across flats	Gage
216	Location of flat	Gage
217	Marking missing, incorrect, or illegible	Visual
218	Radius or chamfer missing, except as otherwise classified	Gage
219	Surface finish improper, except as otherwise classified	Visual

4.5.2.14 Spring, Piston Locking, Drawing 8593871, covering a detail of Drawing 8593855.

Critical
None defined

Major

101	Load at specified height	Gage
102	Height, solid, max	Gage
103	Ends incorrect	Visual
104	Protective finish inadequate or defective	Visual
105	Burr	Visual
106	Foreign matter	Visual

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Minor		
201	Diameter outside, closed, max	Gage
202	Inside diameter of eye	Gage

4.5.2.15 Assembly, Drawing 8593855, less all components except Drawings 8593864, 8593863, and MS28775-012.

Critical		
1	Firing pin protrusion	Gage
Major		
101	Any part missing	Visual
102	Protective finish inadequate or defective	Visual
103	Burr	Visual
104	Foreign matter	Visual
Minor		
None defined		

4.5.2.16 Metal Parts Assembly, Drawing 8593855.

Critical		
None defined		
Major		
101	Trunnion not free to rotate 360 degrees	Manual
102	End sleeve not free to rotate 360 degrees clockwise	Manual
103	Any part missing, except as otherwise classified	Visual
104	Improper assembly	Visual
105	Threads damaged or not full	Visual
106	Burr	Visual
107	Foreign matter	Visual
Minor		
201	Protective cap or plug missing	Visual
202	Protective finish inadequate or defective	Visual

4.5.2.17 Box, Fiberboard (Inner), Packing, Ammunition, Drawing 8797631.

Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Manual
202	Box damaged	Visual
203	Marking missing, incorrect, or illegible	Visual

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204	Cut, damaged, or improperly sealed bag	Visual

4.5.2.18 Box, Packing, Ammunition, Drawing 8797632.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Visual/Manual
202	Strapping missing, loose, broken, or improperly assembled	Visual/Manual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Number of units incorrect	Visual

4.5.2.19 Box, Fiberboard (Outer), Alternative, Packing, Ammunition, Drawing 9221261.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Manual/Visual
202	Carton broken or damaged	Visual
203	Number of units incorrect	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Tape incomplete	Visual

4.5.3 Inspection testing. The following tests shall be performed in accordance with 4.6.

4.5.3.1 Hardness. Fifty units from each heat treated lot of components shall be tested for compliance with the requirements on the applicable drawing. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot of components.

4.5.3.2 Hydrostatic pressure. Each body and end cap shall be subjected to the hydrostatic pressure test. Failure of any cap or chamber subassembly to comply with 3.4 shall be cause for rejection of the subassembly.

4.5.3.3 Shear pin strength. Fifty pins from each lot of shear pins shall be tested for shear strength. Failure of any shear pin to comply with 3.5 shall be cause for rejection of the lot.

4.5.3.4 Protective finish. The number of units for protective finish tests shall be in accordance with the applicable specification. Failure of any unit in the inspection sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

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4.5.3.5 Residual magnetism. Each assembly shall be subjected to the test for excessive residual magnetism as specified in 4.6.5. Failure of any unit in the inspection sample to comply with 3.8 shall be cause for rejection of the assembly.

4.6 Test methods and procedures.

4.6.1 Hardness. Prior to the application of protective finish, a hardness tester shall be used to determine compliance with the requirements of the applicable drawing.

4.6.2 Hydrostatic pressure. Prior to application of protective finish the body and end cap shall be mounted in the device, using a fixture for properly sealing all openings, and subjected to the internal hydrostatic pressure to determine compliance with 3.4.

4.6.3 Shear pin strength. After application of protective finish, the pin shall be subjected to the shear pin strength test to determine compliance with 3.5. The test shall be performed in a fixture in conjunction with a tensile testing machine.

4.6.4 Protective finish. Test procedures for protective finish shall be in accordance with the applicable specification referenced on the applicable drawing.

4.6.5 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end and at the same height as the compass, in a north-south horizontal position (head end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly for four full passes.

4.7 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with the requirements of section 5.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be Level A or B, as specified by the contracting activity (see 6.2). Level A shall be in accordance with Drawings 8797631 and 8797632. Level B shall be in accordance with Drawings 8797631 and 9221261.

5.2 Air Force packaging. Air Force packaging shall be in accordance with TPO 00-891-5488.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The thruster assembly is used in conjunction with other propellant actuated devices in Aircrew Emergency Escape Systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

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a. Title, number, and date of this specification

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2)

c. If first article inspection is required (see 3.1 and 6.3)

d. The responsible inspection activity (see 4.1)

e. Any special instruction requirements (see 4.3)

f. Level of packing and marking (see 5.1)

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors that the items should be a first article sample. The first article should consist of 5 metal parts assemblies, 9 sets of unassembled parts, 1 set of unassembled parts without plating or final protective finish, and 50 shear pins. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definition.

6.4.1 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.5 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.6 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

Custodian:

Preparing activity

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Navy - OS

Navy - OS

Air Force - 99

(Project 1377-0E15)

Review activity:
Air Force - 70

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